6. Find the domain of (i) $f(x)=\sin^{-1}\left(\frac{|x|-2}{3}\right)+\cos^{-1}\left(\frac{1-|x|}{4}\right)$ (ii) $g(x)=\sin^{-1}x+\cos^{-1}x$

7. For what value of x, the inequality $\frac{\pi}{2}<\cos^{-1}(3x-1)<\pi$ holds?

8. Find the value of

(i) $\cos\left(\cos^{-1}\left(\frac{4}{5}\right)+\sin^{-1}\left(\frac{4}{5}\right)\right)$ (ii) $\cos^{-1}\left(\cos\left(\frac{4\pi}{3}\right)\right)+\cos^{-1}\left(\cos\left(\frac{5\pi}{4}\right)\right)$.

4.5 The Tangent Function and the Inverse Tangent Function

We know that the tangent function $y=\tan x$ is used to find heights or distances, such as the height of a building, mountain, or flagpole. The domain of $y=\tan x=\frac{\sin x}{\cos x}$ does not include values of x, which make the denominator zero. So, the tangent function is not defined at $x=\cdots,-\frac{3\pi}{2},-\frac{\pi}{2},\frac{\pi}{2},\frac{3\pi}{2},\cdots$. Thus, the domain of the tangent function $y=\tan x$ is $\left\{x:x\in\mathbb{R},x\neq\frac{\pi}{2}+k\pi,k\in\mathbb{Z}\right\}=\bigcup_{k=-\infty}^{\infty}\left(\frac{2k+1}{2}\pi,\frac{2k+3}{2}\pi\right)$ and the range is $(-\infty,\infty)$.The tangent function $y=\tan x$ has period π.

4.5.1 The graph of tangent function

Graph of the tangent function is useful to find the values of the function over the repeated period of intervals. The tangent function is odd and hence the graph of $y=\tan x$ is symmetric with respect to the origin. Since the period of tangent function is π, we need to determine the graph over some interval of length π. Let us consider the interval $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$ and construct the following table to draw the graph of $y=\tan x$, $x\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$.

x (in radian)	$-\frac{\pi}{3}$	$-\frac{\pi}{4}$	$-\frac{\pi}{6}$	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$
$y=\tan x$	$-\sqrt{3}$	-1	$\frac{-\sqrt{3}}{3}$	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$

Now, plot the points and connect them with a smooth curve for a partial graph of $y=\tan x$, where $-\frac{\pi}{2}\le x\le\frac{\pi}{2}$. If x is close to $\frac{\pi}{2}$ but remains less than $\frac{\pi}{2}$, the $\sin x$ will be close to 1 and $\cos x$ will be positive and close to 0. So, as x approaches to $\frac{\pi}{2}$, the ratio $\frac{\sin x}{\cos x}$ is positive and large and thus approaching to ∞.

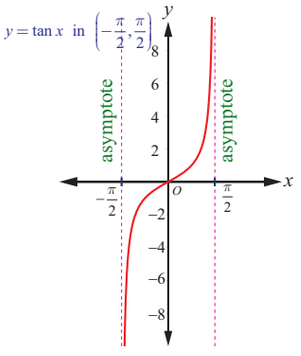


Fig. 4.15

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